

Physics-Guided Visual Air-Quality Estimation

The project explained in full — and what every result means

A from-zero companion to the results. Read this to understand not just the numbers, but what each one actually tells us, what it does not, and what to say about it. Generated 2026-09-28.

1. What this project is (in one breath)

We predict **air quality from a single street photo** — and we do it **honestly**. Instead of spitting out one confident number, the system gives a **low–median–high range**, can **refuse to answer** on unusable photos, estimates **how much error is physically unavoidable**, and is ****evaluated in a way that can't cheat****. The scientific centre of the project is a discovery: the usual way these models are scored **inflates the score through data leakage**, and we measure exactly how much.

The one-sentence version: *estimate air quality from a photo, report it with honest uncertainty, and prove that a popular benchmark's high score is inflated by data leakage.*

2. The units rule (read this first, it trips everyone up)

The label we predict is a **US-EPA Air Quality Index (AQI)** value — a unitless index roughly from **1 to 530**. It is **not** a concentration in micrograms per cubic metre ($\mu\text{g}/\text{m}^3$). Every error number in this document — MAE, RMSE, interval width — is in **AQI points**. (We only touch $\mu\text{g}/\text{m}^3$ inside the error-ceiling analysis, where we convert it to AQI with the official EPA table.)

One more crucial fact about the label: each photo's label is the **daily-average** AQI for that location. A photo is a single instant. Pollution rises and falls during a day, so the label is a slightly "smudged" answer key — a fact that matters a lot in Section 8.

3. How the system works (the pipeline, plainly)

Piece	What it does, in plain words
5-channel input	We feed the network the ordinary photo (Red, Green, Blue) plus two physics maps : a <i>transmission map</i> (how much haze sits between camera and scene, from the Dark Channel Prior) and an <i>inverted-saturation map</i> (haze washes colour out; this catches it, especially in sky where the first map fails).
Backbone (EfficientNet-B0)	A compact, pretrained image network that turns the 5-channel image into features. It is the benchmark's own strongest backbone, so our numbers are comparable.
Interval heads	Three outputs predicting the 5th, 50th, and 95th percentiles — a low, middle, and high estimate. They are built so they can never cross (high $\geq$ middle $\geq$ low, always).
Point head	A separate output trained to hit the average (mean), because the accuracy score R^2 rewards matching the mean. This is what our MAE/RMSE/ R^2 come from.
Conformal calibration	A post-processing step that guarantees the low–high range contains the truth about 90% of the time, for any model.
Isotonic recalibration	A gentle correction, learned on held-out data, that removes systematic under-prediction of high pollution.
Station-grouped split	The evaluation rule: all photos from one monitoring station go entirely into <i>one</i> of train/calibration/test. This is the anti-cheating measure — Section 6 explains why.

4. The measuring sticks — what each metric MEANS

Before any result, here is what each number is actually saying.

- **MAE (Mean Absolute Error)**. The average size of the miss, in AQI points. ****On a typical photo we are off by about this many AQI points.**** Directly interpretable; a MAE of 46 means the typical miss is ~ 46 AQI points. It does **not** depend on how spread out the test set is, which makes it the most honest single accuracy number.

- **RMSE (Root Mean Squared Error).** Like MAE but squares the errors first, so it punishes occasional

big misses more. Always $\geq$ MAE. Useful next to MAE: if RMSE $\gg$ MAE, a few large errors dominate.

- **R² (coefficient of determination).** *What fraction of the variation in the truth does the model*

explain, compared to just always guessing the average?" **1.0** = perfect, **0** = no better than guessing the average, **negative** = worse than guessing the average. It is the strict field-standard score. **Key subtlety:** R² depends on how spread out the test set is (its variance), so the *same model* can score very different R² on different test sets. This is why we always show MAE and the test-set spread next to R².

- **r²(Pearson) (squared correlation).** A looser cousin: it measures only whether the model's ranking

lines up with the truth, ignoring bias and scale. It is always $\geq$ R². Some papers call this "R²." We report **both** so nobody can quote the looser one against us. If someone's "R²" looks suspiciously high, they may be reporting this one.

- **Spearman (rank correlation).** *Does the model put photos in the right order from cleanest to most*

polluted? **1.0** = perfect ordering. A high Spearman with a modest R² means the model sees pollution (ranks it well) but is off on the exact number.

- **Coverage.** For the low–high range: *How often did the truth actually land inside the range?* We

target **0.90**. Coverage is meaningless without width (an infinitely wide range always "covers"), so we always report the two together.

- **Interval width.** The average size of the low–high range, in AQI points. Wider = the model is less

sure.

5. What every honest-accuracy result means

5.1 The headline: cross-validated R² = 0.385 ± 0.19

We evaluated the model on **5 different station-disjoint splits** (5-fold cross-validation) and averaged:

fold	0	1	2	3	4
R ²	0.405	0.574	0.365	0.429	0.152

R² = 0.385 (SD across folds 0.15; 95% CI [0.196, 0.574]) · **MAE = 46.3** · Spearman 0.679 · **coverage = 0.895**.

What it means:

- **0.385** = across genuinely unseen locations, the model explains about **39% of the variation** in

air quality, versus 0% for guessing the average. For predicting pollution from a *single* photo of a *new* place, that is **squarely field-normal** — the closest published analogue (an easier setup) reaches $\approx$ 0.39.

- **MAE 46.3** = the typical miss is about **46 AQI points**. On the 1–530 scale, and given the model is

reading an instant while the label is a daily average, this is a real but expected error.

- **The $\pm$ is a confidence interval, not a spread.** The "95% CI [0.196, 0.574]" has half-width $\approx$ 0.19;

the plain standard deviation across folds is 0.15 (a different quantity — don't confuse them).

5.2 Why the single-split number (0.220) was lower

An earlier single split gave R² = 0.220. Cross-validation showed that split was simply a **hard, high-variance draw** — it happens to be close to the *worst* of the five folds (0.152). **Read 0.220 as the pessimistic end; 0.385 is the robust estimate.** Cross-validation *raised* our honest headline.

5.3 The most important finding hiding in the folds: 0.15–0.57

The five folds range from **0.152 to 0.574**. That huge swing is itself a result: **a single R² on this task is unreliable.** Which stations you happen to hold out changes the score dramatically. This is the deepest reason to distrust any single-number benchmark on this dataset — including the paper's 0.55.

5.4 Coverage = 0.895 — the intervals actually work

Across five independent folds, the truth landed inside the 90% range **89.5%** of the time — almost exactly the 0.90 target. **This validates the honesty machinery:** the calibrated ranges are trustworthy, not just on one lucky split.

Honest caveat. These 5 folds come from one deterministic partition, so they are *correlated* (they share training data). That makes the plain confidence interval a little **optimistic**. A fully

rigorous version repeats the whole thing over several seeds (Nadeau–Bengio correction). The story — 0.385, split-dependent, validated coverage — does not change.

6. What the leakage results mean (the heart of the project)

Data leakage = test information sneaking into training, so the score measures *memorisation* instead of *understanding*. Here the leak is specific: many photos come from the ****same monitoring station and therefore share a near-identical daily-average label****. If such photos land on both sides of the split, the model can score well by **recognising the place**, not by reading the haze.

6.1 The leakage gradient — same model, only the split changes

split	R ²	MAE	SD(y_test)	what it is
random (leaky control)	0.759	26.4	87.4	photos assigned independently → same-station photos on both sides
temporal	0.380	43.5	74.5	train on earlier dates, test on later
shipped	0.378	44.5	81.5	the dataset's own 80/20 split (station-disjoint)
station_grouped (honest)	0.220	54.2	108.1	every station in exactly one split — the honest protocol
geographic (hardest)	−0.079	59.9	100.6	whole regions held out

What it means, row by row:

- **random = 0.759**. Deliberately leaky. It scores **above the benchmark's 0.55**, and above the *entire* honest cross-validated range. A leaky split can manufacture a very high score.
- **temporal / shipped = ~0.38**. Station-disjoint (or nearly), and land near our honest CV mean.
- **station_grouped = 0.220**. The honest single number (the hard draw from Section 5.2).
- **geographic = −0.079**. Holding out whole regions is so hard the model does slightly worse than

guessing — an honest measure of how badly single-photo AQI transfers across the globe.

6.2 The leakage gap (Δ_{leak})

- **Lead with the MAE gap: $\Delta_{\text{leak}}(\text{MAE}) = 54.2 - 26.4 = 27.8$ AQI**. The leaky split roughly ****halves the error****. We lead with MAE because, unlike R², it does not depend on the test set's spread — it is the clean, variance-free measure of the effect.
- $\Delta_{\text{leak}}(\text{R}^2)$ is 0.539 against the single hard draw (0.220), or 0.374 against the robust CV mean

(0.385). We report the pair rather than only the most dramatic number.

6.3 The causal proof — contaminated vs clean

Within the leaky (random) model's own test set, we split photos into **contaminated** (their station or a near-duplicate is also in training) vs **clean** (genuinely unseen):

subset	n	R ²	MAE
contaminated (leaked)	1,757	0.762	25.4
clean (no leakage)	463	−0.86	30.4

What it means: the leaky split's high score lives **entirely in the leaked photos**. On genuinely unseen photos the same model is worse than guessing. This is the clincher: the inflation is **leakage**, not the random test set merely being easier.

Read the −0.86 correctly — it is variance-amplified, not a catastrophe. The clean subset is small (463 photos) and dominated by low-variance stations, so its label spread is small. Because $R^2 = 1 - (\text{RMSE}/\text{SD})^2$, a *modest* error on a low-spread set sends R² sharply negative. Look at the MAE: **25.4 vs 30.4** — the predictions on clean photos are only a *little* worse, not wild. ****Lead with the MAE gap and the direction (contaminated ■ clean); treat −0.86 as qualitative.****

6.4 How to talk about the benchmark's 0.55 (honestly)

We do **not** claim the authors faked anything, and we do **not** claim 0.55 is impossible honestly — one of our honest folds reached **0.574**. What we claim, and prove, is:

1. the data is **dramatically leakage-prone** (a leaky split inflates past 0.55, to 0.759), and
2. single numbers on this task are **unreliable** (honest R^2 swings 0.15–0.57).

The benchmark's unspecified split *admits* the leakage mechanism; our contribution is to **measure and causally prove** the inflation, and to report honestly (cross-validation + calibrated intervals + an error ceiling).

7. What the error-ceiling (C2) result means

Because each label is a **daily average** but each photo is an **instant**, even a *perfect* photo-reader would disagree with the label by the day's within-day swing. That puts a hard **upper bound** on achievable R^2 : $R^2_{\max} = 1 - \text{Var}(\epsilon)/\text{Var}(y)$, where $\text{Var}(\epsilon)$ is the within-day noise.

The result: $R^2_{\max} \approx 0.98$ — measured on 32 OpenAQ stations across 6 countries (1,241 station-days): 0.982 (optimistic) to **0.979** (conservative, imputing the still-uncovered top bands), and ~ 0.98 under both the 2012 and 2024 EPA breakpoint tables. Report it as **~ 0.98 , an upper bound** — not a single decimal, and not with a tight confidence interval (the cluster-bootstrap band [0.970, 0.993] comes from only 32 stations). An earlier 13-station pass gave a looser 0.977, and we worried the true value might fall to ~ 0.90 – 0.95 once the most-polluted bands were covered — the better-sampled re-run shows it does **not**: even imputing the uncovered top $\sim 13\%$ (200+ AQI) with a high within-day variance, the ceiling only moves to 0.979.

What it means for the project: label noise is only **$\sim 2\%$ of the label variance** ($\text{Var}(\epsilon) \approx 208$ – 248 vs $\text{Var}(y) \approx 11,708$). So daily-average labeling costs at most ~ 2 points of R^2 , and our honest 0.22–0.39 is limited by **how hard it is to read pollution from one photo**, not by noisy labels. The ceiling's *job* was to answer "is the gap difficulty or bad labels?" — and the answer is emphatically **difficulty**.

8. Were the numbers checked? Two independent audits

Audit 1 (2026-09-25) and **Audit 2 (2026-09-28)** each used three independent read-only agents to re-read every module and re-check the math. Combined verdict:

- ****The accuracy numbers (0.220 single, 0.385 ± 0.19 CV), the leakage gradient, and the causal**

contaminated/clean result are computed correctly and honestly — no result-affecting bug.** Predictions line up with the right photos and labels; splits are station-disjoint by construction; the coverage guarantee is implemented correctly; the point head genuinely targets the mean; and every table row reconciles with the identity $R^2 \approx 1 - (\text{RMSE}/\text{SD})^2$.

- **The one over-statement was the first-run C2** (a loose 0.977 from 13 stations); the proper re-run

gives $R^2_{\max} \approx 0.98$ (Section 7).

- Everything else was reporting polish, now applied (leading with the MAE gap, flagging the -0.86 as

variance-amplified, disambiguating the confidence interval, noting the correlated-fold caveat), plus defensive safety checks and tests added to the code.

9. What to conclude — and what NOT to conclude

Conclude:

- Honest, robust accuracy is $R^2 \approx 0.385$ with **validated 90% intervals** and $\text{MAE} \approx 46$ AQI.
- The data is **dramatically leakage-prone**: a leaky split inflates to 0.759, proven causally.
- The honest score is limited by the **visual task**, not by label noise (the ceiling).
- Single-number scores on this task are **unreliable** (0.15–0.57 across honest folds).

Do NOT conclude:

- ✗ "0.55 is impossible / the authors cheated." (One honest fold hit 0.574; their split is merely

unspecified and *admits* leakage.)

- ✗ "Clean photos give $R^2 = -0.86$, so the model is broken." (That -0.86 is variance-amplified; the MAE

gap is modest.)

- ✗ "The ceiling is exactly 0.98 / 0.977." (It is an upper bound ~ 0.98 with the top $\sim 13\%$ extrapolated;

report it as ~ 0.98 , not a single decimal, and not with a tight CI.)

- ✗ " R^2 alone tells the story." (Always read it beside MAE and the test-set spread.)

10. What still needs running (short list)

Done: core pipeline, leakage, cross-validation, and the **C2 error ceiling** (≈ 0.98 , Section 7). C3 abstention and the demo are **built and locally tested** — they just need one run each:

1. **08_abstention.ipynb** on Kaggle — attach ExDark (night) / DTD (textures) / MIT-Indoor (indoor),

run it (uses the trained model, no retrain). Produces the risk–coverage/AURC + OOD tables and saves the demo bundle. *This is the one worth doing next.*

1. **Deploy the demo** — download the bundle, then follow **docs/11_demo.md** to create the free Hugging

Face Space (Gradio) and go live at a permanent URL.

1. **(Optional) 10_crossval.ipynb with ACCURACY_PUSH = True** — ensemble + EMA + TTA to try to raise

R^2 (~ 15 h on a GPU); reported as a *new row* beside the baseline. Skip unless you want a higher number.

11. Mini-glossary

- **AQI** — Air Quality Index, a unitless 1–530 index (what we predict). Not $\mu\text{g}/\text{m}^3$.
- **Leakage** — test info reaching training, so the score measures memorisation, not understanding.
- **Station-grouped split** — all of one station's photos go to one split; the honest protocol.
- **Cross-validation (GroupKFold)** — rotate which fifth of the *stations* is the test set, five times;

report mean $\pm$ CI.

- **R^2 / MAE / RMSE / Spearman / coverage** — see Section 4.
- **Conformal calibration** — the procedure that guarantees $\sim 90\%$ interval coverage for any model.
- **Δ_{leak}** — how much a leaky split inflates the score over the honest one.
- **R^2_{max} (error ceiling)** — the best R^2 any model could reach given the daily-average labels (an

upper bound, measured ≈ 0.98).